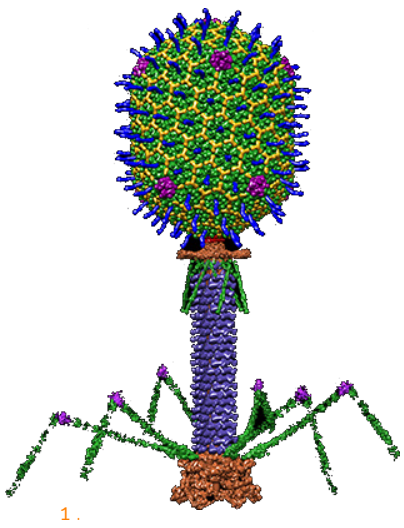
BACTERIOPHAGE

MORPHOLOGY

Phage have a highly distinctive **geometric design.**



CAPSID contains DNA

BARREL head-to-tail interface

TAIL ejects DNA

LEGS adsorption apparatus

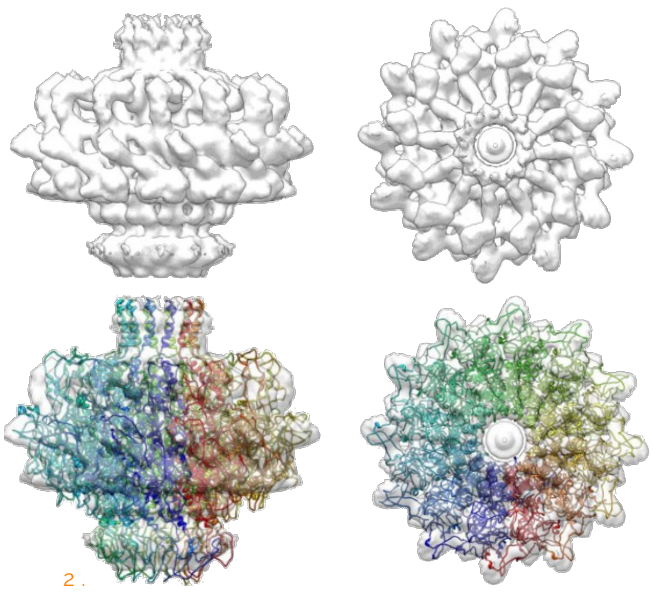
1.

Phage taxonomy is diverse but the best studied phage have **double- stranded DNA** and are of the order **Caudovirales.**

Their **icosahedral** capsid has 20 faces.

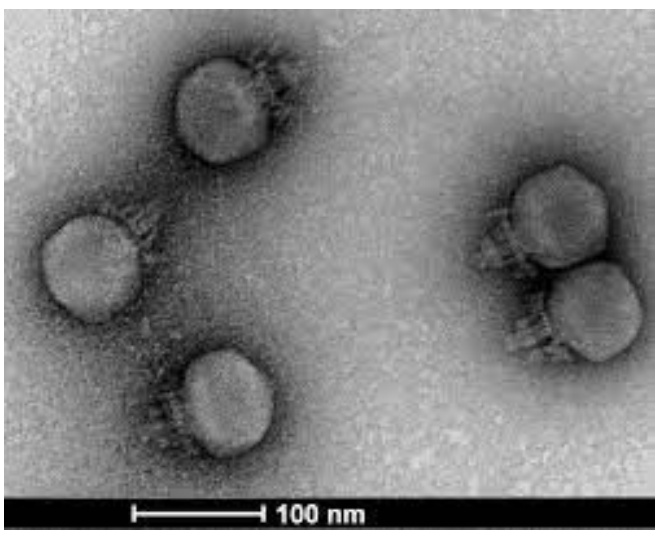
X-ray crystallography resolved the first structure of the phage capsid at 3.6 Å (angstrom).

The portal protein is located at the **head-to-tail interface.**



2.

Dodacameric (12 proteins) in structure, the **portal protein** forms a vertex valve which controls the release of genetic material from the capsid upon phage binding with the host bacterial cell receptor.



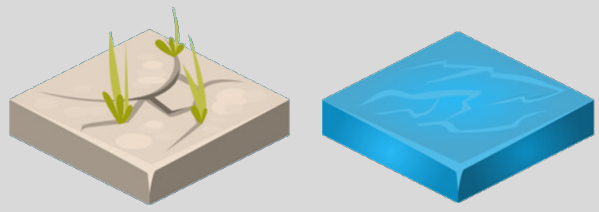
CrAss001, a member of the most abundant bacteriophage family in the human gut, crAssphage.[3]

3 billion years of evolution have uniquely shaped phage for maximum efficiency at their sole purpose - replication.

As **obligate parasites**, to survive and reproduce phage DNA must enter a host bacterium.

ABUNDANCE

Phage are the most abundant organisms on Earth!



10^8/g in soil **10^10/L** in ocean

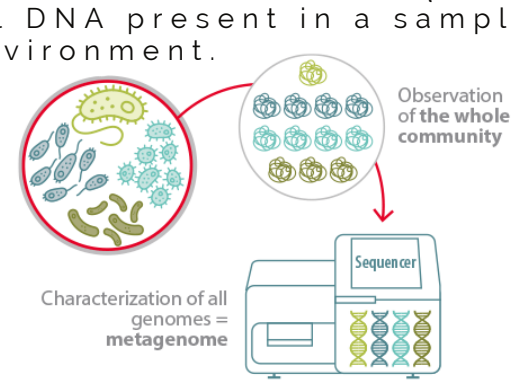
Found in every explored biome, from the human gut to the deep ocean, soil and ancient stool specimens.

Keen bacterial snipers, phage play a crucial role in the ecosystems they occupy.

In the ocean, phage are involved in **biogeochemical recycling** through a process called the **viral shunt.**[4]

VIRAL DARK MATTER

Metagenomics sequencing is a revolutionary technology which can detect and quantify all DNA present in a sampled environment.



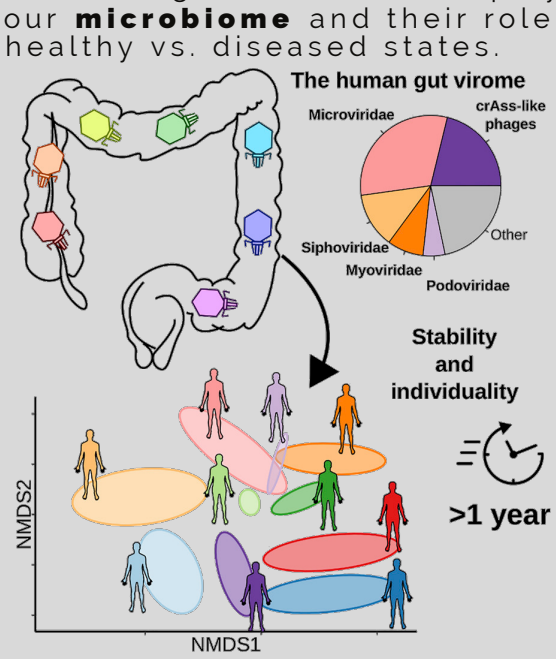
Most known viral sequences originate from metagenomics.

Up to **90%** of these detected sequences can have **no known homology** to existing reference genomes and have been termed **viral dark matter.**[5]

This leaves great potential for the discovery and classification of new viruses.

HUMAN GUT PHAGEOME

Millions of viruses live in harmony in our gut. Fascinating new research is uncovering the role viruses play in our **microbiome** and their role in healthy vs. diseased states.

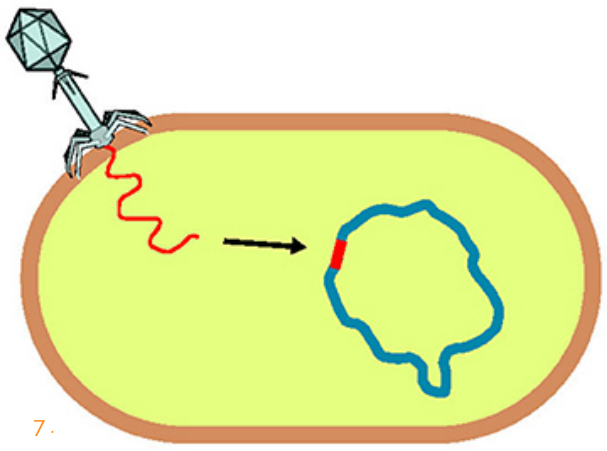


Virome composition is **highly stable** and **unique** to each individual.[6]

LIFE CYCLE

Some bacteriophage are double agents. They can switch between **dormant** and **active** replication states.

LYSOGENIC (DORMANT)



7.

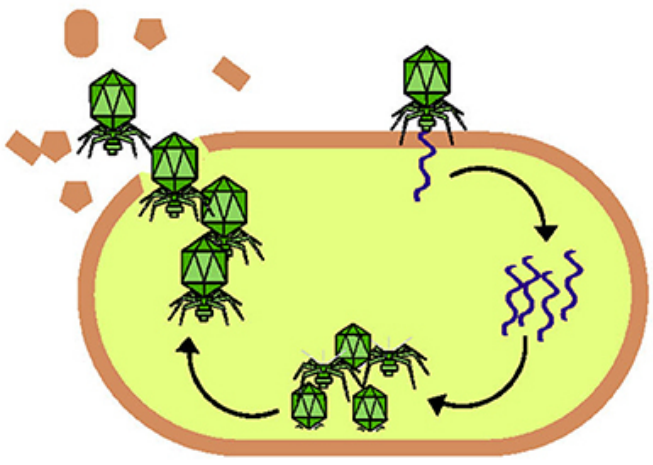
Phage incorporate their DNA into a host bacteria's chromosomal DNA to become a **prophage.**

The prophage quietly exists in incognito mode piggybacking off the cell's resources.

Prophage DNA can become expressed spontaneously or upon detecting changes in environmental conditions such as cell stress or DNA damage.

This kick starts the lytic cycle.

LYTIC (VIRULENT)



The bacterial cell is converted into a factory for the production and assembly of phage machinery.

This **virulent** state is most associated with viruses. An army of phage burst from the host to invade nearby bacteria.

FOOD IMPACTS STATE

Research has shown different foods and chemicals can stimulate prophage **induction.**[8]

Bacteria treated with stevia and bee propolis induced the largest numbers of phage while rhubarb, coffee and oregano significantly inhibited induction.

TRANSDUCTION

An interesting phenomenon is the packaging and shipping of bacterial DNA by phage to other bacteria in the community via **generalized** or **lateral** transduction.

This complex phage-host relationship is thought to drive microbial evolution and increase bacterial fitness through gene shuffling by conferring traits e.g. **antimicrobial resistance.**[9]

1. https://www.drvictorpadillasanchez.com/Images.htm
2. Lokareddy, Ravi K., et al. "Portal protein functions akin to a DNA-sensor that couples genome-packaging to icosahedral capsid maturation." Nature communications 8.1 (2017): 1-11.
3. Shkoporov, Andrey N., et al. "CrAss001, a member of the most abundant bacteriophage family in the human gut, infects Bacteroides." bioRxiv (2018): 354837.
4. Breitbart, Mya, et al. "Phage puppet masters of the marine microbial realm." Nature Microbiology 3.7 (2018): 754-766.
5. Roux, Simon, et al. "Viral dark matter and virus–host interactions resolved from publicly available microbial genomes." elife 4 (2015): e08490.
6. Shkoporov, Andrey N., et al. "The human gut virome is highly diverse, stable, and individual specific." Cell host & microbe 26.4 (2019): 527-541.
7. Guerin, Emma, and Colin Hill. "Shining light on human gut bacteriophages. Frontiers in cellular and infection microbiology 10 (2020).
8. Boling, Lance, et al. "Dietary prophage inducers and antimicrobials: toward landscaping the human gut microbiome." Gut microbes 11.4 (2020): 721-734.
9. Fillol-Salom, Alfred, et al. "Bacteriophages benefit from generalized transduction." PLoS pathogens 15.7 (2019): e1007888.